

# Mobile Technology Trends in Higher Education

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This paper examines current information on mobile technology trends and how these technologies, as well as podcasting, are utilized in teaching and learning in today's information age and higher education. Five current technology trends are discussed to address the benefits of today's technological changes and lifestyles and learning styles of students: (a) technology and social network communities, (b) the impact of technology on higher education, (c) the impact of technology to impart learning, (d) the use of podcasting for instruction, and (e) learning theories, practices, and earlier studies in the area of technology-mediated teaching and learning. Mobile devices (MP3 players, MP4 players, cell phones, PDAs, and related media devices), as well as podcasting, have emerged as the preeminent tools for teaching and learning in most academic institutions. Without a thorough understanding of current technology trends, it will be difficult to justify the necessary balance between podcasting and traditional classroom instruction.

This paper shows that by leveraging podcasting in the classroom, students are motivated, engaged in the learning process, and more likely to perform better and participate effectively in collaborative coursework. The philosophical approach to education through podcasting is to distribute classroom materials to students via personal computers and mobile devices. It is not the intention of this paper to change or replace traditional forms of teaching and learning, but rather to encourage podcasting as an alternative for relaying classroom instructions or as supplemental

instruction for disadvantaged students. The approaches utilize mobile technologies and foment outreach by adapting to the technological culture of the student population.

Current trends in mobile and podcasting technologies have gained attention for their ability to distribute audio/video on demand for playback. Mobile and podcasting technologies are relatively new movements in higher education and have already been woven into the social fabric (Kim, Mims, & Holmes, 2006). Mobile technologies are being widely adopted by mainstream news media outlets, social network communities, bloggers, and information exchanges. These formats are now becoming the foremost technologies in the education market and are particularly very popular among young people, including college students (Fini, 2008, p. 714). Altbach, Berdahl, and Gumport (2005) argued that “students' consumerism is a central part of the ethos of American higher education” (p. 307). Altbach noted that education in the United States has focused primarily on students and their lives. Based on these assumptions, American higher education institutions will generally be more likely to embrace students' cultures to bring about educational reform. Podcasting is that new technological culture.

Pedagogical approaches based on these assumptions essentially create alternative learning for gathering, customizing, and sharing information among teachers and students, enabling them to reach out to other students in disparate localities (Kidd, 2008, p. 847; Vequist, 2008, pp. 87-88). As a result of mobile and podcasting technologies, the academic environment has shifted from traditional classroom settings to mobile learning centers. A study by the Pew Internet and American Life Project (2008) found that 96% of Americans aged 18 to 29 reported owning a cell phone or PDA. Innovation in mobile technologies has enabled the

distribution of audio and video podcasts for playback on mobile devices (Sorrentino, 2008c, p. 512).

The art of mobile-centric and podcasting technologies has emerged as new educational tools gaining widespread acceptance in higher education (Kim et al., 2006). Podcasting allows students and educators to communicate, interact, and collaborate in classroom instruction in more meaningful ways (Jham et al., 2008). This pedagogical practice consists of traditional forms of classroom education, in which students attend lectures, take notes, and review reference materials on their cell phones, PDAs, MP3 players, or MP4 players. Instead of focusing on the lecturers while taking notes in the classroom, students can schedule in-person attendance or participate in lecture discussions rather than focusing on notetaking. This method allows students and teachers to banter in a more interactive and collaborative learning environment while still being able to review classroom podcasts they have subscribed to on their mobile devices or other related media players. The power of mobile centric and podcasting technologies, combined with pervasive access to the Internet and Web technologies, delivers distance education in American colleges and universities, European schools, and advanced academic institutions around the world, thereby creating new avenues to deliver flexible and boundless learning in today's culture of abundant streaming technologies and captive instructional styles (Inoue, 2008, p. 17; Ng, 2008, p. 21; Ololube, 2008, p. 100).

### **Technology and Society**

Technological advances of the 1990s and the broad accessibility of the Internet have profoundly transformed traditional society into an information society. These technological advances and innovations have transformed every aspect of society,

especially in business, higher education, government, and the global industrial complex (Sorrentino, 2008a, p. 32). Some widely used terms describing the abundance of information and knowledge databases accessible in the information society are the knowledge society, the Internet society, the digital society, and now the *mobile-centered society*. Information and communication technologies remain the key factors in any organizational development, particularly in higher education. By incorporating state-of-the-art technologies into their operations, organizations can function more effectively and efficiently and outperform competitors (Chen, 2008, p. 401). Academic institutions are not-for-profit businesses; however, having these technologies enables them to provide better tools for communication and collaboration, as well as for teaching and learning. Many higher education institutions worldwide have already adopted advanced technologies to improve communication and collaboration among stakeholders and in teaching and learning (Inoue, 2008, p. 14).

Over the past two decades, advances in communication technologies, such as faster computing power and advanced Internet capabilities, have become the driving force behind enabling people worldwide to communicate and exchange information quickly (Cartelli & Palma, 2008, p. xiii). Businesses, government, higher education, and global industries have utilized information and communication technology to streamline operations, function more effectively, and become more efficient (Ololube, 2008, p. 100). Advances in computer technology, computer networks, computer software and applications, and the Internet have enabled these organizations to communicate with their stakeholders more effectively and efficiently. With the Internet as a centralized network, people throughout the world can send and receive

information to anyone, and that information can be used at home, at work, and around the world.

Today, these computer services are widely available on personal computers, handheld devices (PDAs), and mobile devices that make it possible for individuals and organizations to interact and exchange information. These technologies are widely used to enhance students' learning experiences in higher education. Similarly, teachers, learners, and parents can participate and interact together in a learning community (Hussain, 2008, p. 72). The daily lives and lifestyles of every person in the information society have been greatly improved by the advent of personal computers, the Internet, and most recently, mobile technology. Increasingly, mobile devices have become the common means for people to exchange information from anywhere in the world. Emerging mobile technologies are considered to be primary, and their advances will change the way people live (Kunz & Gaddah, 2008, p. 25).

According to the Pew Internet and American Life Project (2009a), Internet and mobile phone users tend to have larger and more diverse networks of close confidants than do those who do not use the Internet or mobile phone. They found that the use of social networking services, particularly Facebook and Twitter, was more strongly associated with more diverse social networks. The reality of the information society is that new communication technologies enable and even encourage people to interact with others. As social networking and Web technology have enabled new ways of communicating, structuring, and sharing knowledge, podcasting mediates between learners and knowledge. Multimedia (audio, video, graphics, and text) can be combined to deliver an interactive learning environment that encourages "individuals to learn together with others, maintaining their control over time, space, presence,

activities, identity and relationships” (Anderson, 2005, as cited in Fini, 2008, p. 713). Podcasting is a new multimedia-based instruction.

Similar to the information society, advanced technologies are also transforming the global economy; emerging mobile technologies have brought significant changes in the information age to the transformation of global society. In recent years, mobile devices have become much more technically sophisticated. Mobile devices such as iPhones, smartphones, and PDAs have integrated wireless capabilities that allow students to access the Internet and perform various network-related functions (Kidd, 2008, pp. 848-850). Increasingly, these devices are becoming multipurpose tools for daily tasks, especially for personal, business, and educational activities (Kunz & Gaddah, 2008, p. 25). The daily lives of everyone are increasingly dependent on mobile devices for communication, entertainment, information, and managing everyday needs (schedules, contacts, etc.). Mobile technologies have brought about rapid changes in today’s higher education and the larger society. Gartner (2008) found that people who used mobile devices were more efficient and productive than those who did not. They predicted that, by 2012, one quarter of the world population would be using mobile devices. All signs point to mobile computing having an even greater footprint in the future of higher education while becoming more entrenched in everyday life (Sorrentino, 2008a, p. 36). This is essentially what Francis Parker (1837-1902), a father of U.S. progressivism, implied: Learners should have direct experience with objects in the environment. Mobile computing and other technological advances have transformed the art of teaching and learning (Gutek, 1997, p. 70).

To prepare for the future information society, and in collaboration with the International Telecommunication Union (ITU), the United Nations (UN) organized the

first World Summit on the Information Society (WSIS) to provide guidance on applying information and communication technologies to benefit humanity. The first principle of the Geneva declaration stated:

We, the representatives of the peoples of the world, assembled in Geneva from 10-12 December 2003 for the first phase of the World Summit on the Information Society, declare our common desire and commitment to build a people-centred, inclusive and development-oriented Information Society, where everyone can create, access, utilize and share information and knowledge, enabling individuals, communities and peoples to achieve their full potential in promoting their sustainable development and improving their quality of life, premised on the purposes and principles of the Charter of the United Nations and respecting fully and upholding the Universal Declaration of Human Rights. (WSIS, 2003)

### **The Impact of Technology on Higher Education**

Since the 1990s, Internet and Web technologies have revolutionized world commerce and communication. The WWW has reached a critical mass that touches the lives of every individual and business. These technologies have also greatly altered the academic landscape of higher education (Ololube, 2008, p. 100). The Internet and Web technologies have created opportunities for information exchange, technology-facilitated teaching and learning, and distance learning in higher education. Based on the ideas of Readings (1996, as cited in Robins & Webster, 2002), Robins and Webster argued that technological changes have transformed the “university of excellence, alias the technological university or the corporate university” (p. 4).

Today’s higher education institutions worldwide offer academic programs in various distinctive forms: (a) classroom education, (b) distance education, (c) online education, and (d) hybrid education (Cartelli, 2008a, p. 350). Like the Internet and Web technologies, emerging mobile technology adds new dimensions for students to learn, communicate, and exchange information. A study by the Pew Internet and

American Life Project (2008) found that large percentages of college-age adults (86% of those ages 30 to 49 years, 64% of those ages 50 to 64 years, and 36% of those ages 65 and over) reported owning a cell phone or PDA.

College students are increasingly dependent on mobile technologies for learning, communication, entertainment, and exchanging information. As Internet and Web technologies have helped businesses and industries worldwide to operate more effectively and efficiently, mobile technologies are similarly having a revolutionary impact on the educational system (Chen, 2008, p 401; Sorrentino, 2008a, p. 39). Research has shown that American universities, European schools, and academic institutions around the world are increasingly adopting the same kinds of technologies used in business enterprises for teaching and learning, as well as for operational efficiency, communication, and information dissemination (Raisinghani, 2008, pp. 526-528). Classroom instructional environments have also blended these technologies to facilitate, publish, and distribute learning materials. Some commonly used technologies in classroom education include computers, projectors, presentation software, the Internet, faculty websites, email, and Web-based course tools (Lee, Pillay, & Chandra, 2008, pp. 532-535).

A study by the University Center for Research, Teaching Innovation and Distance Learning (CARID) found that technology-enabled learning was increasingly integrated with and interacting more with knowledge databases (Poletti, 2008, p. 96). Findings suggested that technology focuses on each individual person to manage his/her learning needs. A study by UCLA found that personal knowledge management concentrated on five main tasks: (a) searching/finding, (b) categorizing/ classifying, (c) naming things/making distinctions, (d) evaluating/assessing, and (e) integrating/relating (Sorrentino, 2008c, p. 512). These two principles suggest that

individual students or educators are responsible for planning and mapping out their own teaching/learning experiences. These principles are consistent with self-directed learning, constructivism, and transformational theories, which suggest that individuals take ownership of planning, carrying out, and evaluating their own learning experiences (Merriam & Cafarella, 1999, pp. 318-320). Gallerani (2008) also aligned his theory of self-constructed autonomy (p. 520). The self-constructed autonomy defines the meaning of the free explanation of each individual's training path, and the constructed dependence defines the other side: the indissoluble/relentless rooting in a net of relations within a community. This also suggests that learning is considered in terms of both the social and cultural aspects of society.

These technologies have been accepted as indispensable parts of industrialized societies. They must further adapt to meet the challenges of the knowledge age (Corazza, 2008, pp. 357-360). American higher education has traditionally responded to changing student curricular interests by expanding fields and departments and cutting unpopular areas; thus, students' interests can affect academic policy and governance (Altbach et al., 2005, p. 307). Adapting to technological change is a natural process of human behavior.

The future technology trends are expected to grow even more rapidly than the Internet and will continue to shape the nature of the workplace and higher education. Many scholars in the area of educational research believe that future trends of the education system in the 21st century are linked to technological advances (Inoue, 2008, p. 17; Vanin et al., 2008, p. 3). As Poletti (2008) pointed out, technology is "a new reality" in higher education where methodological practice can be used in a variety of public and private sectors, ranging from school and university to professional and corporate training. Poletti suggested that the world of education and

student learning training is becoming a new way of structuring and sharing knowledge, and of shaping the culture from which such knowledge derives, and that it must be offered to the society of knowledge (pp. 95-96). This suggests that technology is a structured learning process that encourages a culture of sharing and the acquisition of new knowledge.

A 2003 study by Boettcher (as cited in Lee et al., 2008) focused on online learning systems. Boettcher argued that the structure of the learning content is vital to its success. She listed four key core components of the learning experience: (a) the course material to be learned, (b) teaching, (c) the learner, and (d) the environment. The course material to be learned is only one of the four components of learning. This is essentially what podcasting is about. Podcasting refers to how course content is packaged and delivered to students. The other three components are also related to podcasting as teaching and learning tools. The environment component is related to the flexible learning environment. Boettcher suggested three ways to design course content: (a) Content must be semantically well structured for instruction, (b) content must be a good fit or well structured for a particular student, and (c) content must be technologically well structured. The first way corresponds to the teaching component of the learning experience; the second, to the learner component; and the third, to the environmental component.

Podcasting is technology-based learning. It is a structured learning process in which course content is tailored using all of Boettcher's three ways. These findings support the use of podcasting as a teaching and learning tool and illuminate why podcasting is a "good fit" for today's technological environment (Lee et al., 2008, pp. 532-533). Boettcher's study was also supported by American psychologist Jerome Bruner's research on how students understood "the conceptual underpinnings of

knowledge.” Bruner argued that learning depends on environmental and experiential factors. “The way in which instruction was organized and what was emphasized during instruction were keys to improving learning” (as cited in Salkind & Rasmussen, 2008, p. 964).

### **The Impact of Technology on Teaching and Learning**

According to Gumpert and Chun (2005), there are three potential influences of technology on higher education: (a) the nature of knowledge, (b) the process of teaching and learning, and (c) the social organization of teaching and learning in higher education. The nature of knowledge is determined by how people are involved in its production and by how academic knowledge is valued and disseminated. Teaching is the process in which teachers and students come together in the same place at the same time to disseminate knowledge. The third potential influence is the social organization of teaching and learning in higher education, as determined by the expansion of higher education delivery (pp. 402-412). This idea suggests that technology can be used to extend the traditional teaching and learning process without altering the underlying traditional forms of education.

Traditionally, teachers and students come together in a classroom setting to disseminate knowledge. Based on the work of Cornford and Pollock, “Working through the Work of Making Work Mobile,” Robins and Webster (2002) stated that “the emergence of commercially-oriented, digital, online or virtual university” allowed higher educational institutions to create new and different ways of producing, distributing, and consuming knowledge that learning does not have to be “campus-based” (p. 87). With educational podcasting, knowledge is packaged and information is exchanged between teachers and students; teachers are the producers and students

are the consumers of knowledge. Teachers and students do not have to be at the same place at the same time, where a physical classroom is a virtual place for teaching and learning (Zajda & Gibbs, 2009, p. 21).

Well-known technologist Bill Gates spoke about “the Web lifestyle” as a catalyst for change in collaborative learning communities, in which the Internet allows teachers and students to explore subject matter in new ways and for learning not to be confined to a classroom (Gates, 1999, p. 387). Today’s teaching and learning are no longer restricted to teachers and students coming together in the same place at the same time, and a physical classroom is not the only place for teaching and learning; they can take place in both traditional classroom education and distance education (Manca, 2008, pp. 121-122). Like the Internet and Web technologies, which create opportunities for distance education, e-learning and online education provide anywhere, anytime learning for students without the constraints of a physical classroom.

Many American universities and academic institutions around the world have utilized technology and Web-mediated learning for academic training. Classroom education has blended technology to facilitate teaching and learning through computers, projectors, presentation software, the Internet, multimedia, and Web-based course tools in classroom learning environments (Ng, 2008, pp. 21-29). Current trends in mobile technology have expanded teaching and learning beyond traditional classroom and distance education from flexible practices.

Distance education, such as online and e-learning, uses the Internet and Web-mediated technologies to facilitate traditional classroom learning. Academic institutions around the world have adopted these technologies to create distance learning opportunities for their students and to promote interaction among students

and between teachers and students (Ng, 2008, p. 21). Several popular Web-based course management systems for students include Blackboard, TopClass, and government-sponsored Moodle. These learning systems used Web-mediated technologies to provide a learning environment that is alternative to traditional classroom settings. In addition, these learning systems allow flexible learning environments for students, such as study from home, at their own pace, and while they maintain a personal life or job. The International Technology Education Association (ISTE) defined educational technology as “using multimedia technologies or audiovisual aids as a tool to enhance the teaching and learning process” (Salkind & Rasmussen, 2008, p. 314). These learning systems are closely aligned with ISTE. According to Driscoll (1998, as cited in Donnelly & McSweeney, 2008, pp. 51-52), a Web-based learning platform provides a complete analysis of all types of asynchronous interactions between teachers and students, students and students, and students and course material. Current technological trends in mobile learning have created flexible learning methods for teachers and students to effectively communicate, interact, and engage in classroom learning activities that complement traditional classroom and distance education.

Traditional classroom and distance education, and now mobile teaching and learning have created new opportunities for mobile learning or m-learning (Kim et al., 2006). Business, education, and people worldwide have increasingly relied on mobile technology for everyday purposes. As the Internet has dominated the world of information, mobile technology is a new way to access the Internet. Mobile devices such as cell phones, PDAs, and related mobile devices are much more than just handheld computer systems; they are multi-purpose devices for communication, accessing the Internet, PDA, and educational purposes. In recent years, innovations in

mobile technologies have enabled mobile devices to access distance learning, distributed learning, e-learning, and now m-learning (Frignani, 2008, p. 171). These devices create new opportunities for flexible learning in addition to communication, collaboration, and exchange information (Kidd, 2008, pp. 847-851). In this context, mobile technology has reshaped classroom learning and distance learning. Mobile learning has given a new meaning to current distance education.

The pedagogical approaches to podcast-mobile-assisted learning combine both information technology and traditional classroom teaching and learning to provide the same learning experience as traditional classroom education. Commonly used online discussion groups, videoconferencing, and online resources in distance education are now incorporated in m-learning (Hussain, 2008, p. 69; Reid-Martinez et al., 2008, p. 703; Vanin et al., 2008, p. 1). Several studies have shown that teachers have more positive attitudes toward the potential of technology-based education and teachers who lacked technology experience had negative attitudes toward computers and did not use technology in their teaching (Jimoyiannis, 2008b, p. 325). For this reason, the future of higher education must be directed to the training of teachers in adapting to the new learning environment and becoming more integrated with technology.

### **The Use of Podcasting for Teaching and Learning**

As Internet and Web technologies have created distance education and online learning environments, innovations in podcasting have added a new dimension of mobile learning to traditional classroom and distance education (Song & Kidd, 2008, p. 569). Mobile devices such as cell phones, PDAs, and related media devices have enabled the delivery of classroom education through podcasting. Podcasting is simply the process of packaging classroom learning activities into audio/video podcast files

and distributing them over the Internet to students' mobile devices for playback. The packaging process is the recording of classroom learning activities into audio/video podcast (or MP3/MP4) files and posting them on faculty Web sites or Apple's iTunes software application. Learners access, download, and playback audio/video podcasts of the learning activities that take place in the classroom. Learners access and review lessons covered in the classroom, listening to them once or multiple times. With audio/video podcasts, learners are in total control of their own learning, without time or geographic constraints. Unlike distance learning, podcasting is a prerecording of audio/video podcast of classroom learning activities. Like distance learning, podcasting learning is a flexible learning process beyond a physical classroom and geographic location, and the learners still have the same classroom learning experience. Jonassen (2006) described that the role of technology is to "communicate ideas to learners" (p. xiii). Podcasting is essentially a teaching/learning tool that has the capacity to deliver classroom learning activities via mobile devices.

Podcasting is a supplement to classroom instruction, essentially a recording of class lectures and reviewing them outside the classroom. This approach is not a new concept, but podcasting of class lectures via mobile devices for students at the community college is a new concept. According to Salkind and Rasmussen (2008), educational technology started around the 1950s and focused primarily on the use of instructional media. However, the use of audio recordings for listening started as early as the 1920s (p. 314). An exploratory study using tape-recorded journals by Aitken and Deaker (2007) concluded that the use of tape recording was an effective means of expression for some students and an important aid to critical thinking. Today's podcast has emerged as a new form of tape recording and is evolving into a new phase of educational technology.

Before exploring the benefits of podcasting in educational settings or its acceptance as a teaching/learning tool, it is prudent to first understand what it is. In this way, actions can be taken to determine whether podcasting is an effective teaching/ learning tool for today's technological society. The term *podcasting* is derived from Apple's portable media players, iPods. Podcasting technologies have the capability to package and deliver multimedia content in podcast file format (or MP3/MP4). Learners can subscribe, access, download, and play back audio/video podcasts on iPods, MP3/MP4 players, iTunes (Apple's media software player), Windows Media Player, or related mobile devices.

In technical terms, podcasting is an audio/video files feed, in Really Simple Syndication (RSS) feed format, which is a family of Web feed formats used to publish frequently updated digital content. The term *podcasting* is also used to describe a collection of technologies for automatically distributing audio/video content over the Internet. Educational podcasting can be described as the recording and distribution of the instructional materials such as lectures, class activities, and assignments in digital format or MP3/MP4 format. Windows Media Player, iTunes software application, or related media players can play back audio/video podcasts. However, the term *podcast* still refers largely to audio/video content distribution. Audio/video podcast is either in "push and pull" or audio/video files feed through RSS. A "push and pull" is essentially an audio/video podcast in MP3/MP4 that users can download. An audio/video files feed is automatically synchronized with the users' mobile devices.

In educational terms, podcasting is simply a process for teachers to create self-published course content and distribute it as podcast files. Educational strategies have two aims: (a) enhancing the method of teaching, and (b) helping students to learn more effectively. First, the teacher can improve teaching and presentation skills by listening

to their own lectures and classroom activities. Second, students can download and play back any published classroom podcasts for the lessons covered in class. Students can listen to or review any of the lectures or related learning lessons outside the classroom. The pedagogical model is a “listen and learn” approach that allows students to play back audio/video podcasts of the lessons on their mobile devices. With podcasting, students may choose to listen to the same lessons once or multiple times. When students actively engage in the lesson by listening to the same podcast multiple times, they construct knowledge and begin to crystallize the information they have accumulated. This idea is similar to the work of Herasymowych on increasing intelligence through actions that reflect on the learning cycle, action learning, and learnable intelligence (Merriam & Cafarella, 1999, pp. 188-189).

Mobile technologies and podcasting are tools to help students to think more effectively. As a teaching tool, podcasting works like learning lessons as tutorials on mobile devices that provide step-by-step instructions to help students to construct the meaning of the learning lessons. Teachers provide lessons through audio/video podcasts that play on students’ iPods, MP3/MP4 players, cell phones, or other media players. Traditionally, learners attend lectures to get clarification on learning topics. This teaching method builds on the success of the Internet and Web technologies, aiming to create new opportunities for faculty and students to collaborate and engage in classroom teaching in new ways, enabling faculty to create and deliver course materials as audio/video podcasts. By combining traditional classroom methods with the Internet and mobile devices, teachers can create effective lessons such as audio/video podcasts. According to Kim et al. (2006), learning through mobile devices is a new form of m-learning with no limitations on place or time.

The learning process of podcasting uses the Internet to download audio/video podcasts via “push and pull” or RSS feeds, and to play them on students’ mobile devices or related media players. This approach combines both classroom and distance learning (i.e., online and e-learning) to create m-learning. As Pieri and Diamantini (2008) have stated, e-learning is “the natural evolution” (p. 548). This pedagogical approach to podcasting will not replace the traditional classroom experience; it simply reinforces classroom learning and offers students the flexibility to learn in a technological culture with which they are comfortable. This approach utilizes mobile technology and the Internet to distribute classroom education to mobile devices. Podcasting offers a flexible way to learn beyond traditional classroom education. This form of education is currently very appealing to working students because podcasting provides flexible learning that fits students' learning preferences and culture (Inoue, 2008, p. 19). Podcasting aims to provide students with additional time to review previously covered classroom learning activities and maintain the same learning experience.

### **Learning Theories and Practices**

The art of teaching is the shaping of individual lives to influence their integration into productive societies; it requires blending theory and practice. Practice is guided by proven theories (Gutek, 1997, p. 1). A philosophy of education is simple: to help a person reliably understand the social tenets of reality, the essence of truth, and values.

There are essentially three branches of philosophy: metaphysics, epistemology, and axiology. Metaphysics deals with social realities. Epistemology is the study of the essence of truth and the theory of knowledge. Axiology is concerned with value-based

theories (Gutek, 1997, pp. 2-3). These fundamental branches of philosophy help to determine whether the use of mobile-centric and podcasting devices advances the field of education significantly or materially. Three basic concerns are addressed here: (a) theorizing the practice of mobile and podcasting technologies, (b) the theory of knowing and knowledge society, and (c) the societal value of mobile and podcasting technologies.

Based on educational theories of learning and the nature of reality, the pedagogical practice of podcasting combines constructivism, behaviorism, social learning theory, and cognitive theory. Behaviorism presupposes that students actively engage in self-directed learning, social learning theory presupposes that students learn through the help of audio/video podcasts, and cognitive theory presupposes that students learn through building on prior experiences and through meaningful and relevant experiences (Merriam & Cafarella, 1999, pp. 263-266; Wong & Kerr, 2008, p. 40). Constructivist learning theory suggests that students learn when they actively construct knowledge from listening to classroom podcasts (p. 182). Social learning theory is considered to be one of the most influential theories in the fields of education and psychology. Social learning theory in terms of human functioning and personality development is linked to culture, society, and community. The theory is based on cognitive, social-interactive, self-regulatory, and self-reflective capabilities and processes and concludes that social network communities have a great influence on human learning and development (p. 919). In addition to the impact of social interactions, social learning theorists believe that media have a great influence on human learning and development. These theories suggest that teachers and students are closely related to cultural values and learning styles (Salkind & Rasmussen, 2008, p. 152). Based on these theories, mobile and podcasting technologies can be used to

facilitate cultural sharing, embrace technological society, and enable teachers and students to collaborate. The relationship between theory and practice is about innovating ways to adapt to specific learning situations. The nature of reality is that society embraces technological evolution that drives innovation and new ideas. According to Smith, the nature of reality and the essence of truth are “true to experience” and reflect the “real world” (2003, as cited in Curren, 2003, p. 384).

Podcasting learning can be based on perceptual development. It consists of constructivism, behaviorism, social learning theory, and cognitive theory. Salkind and Rasmussen (2008) mentioned two perceptual developments of senses: sensations and perceptions. Sensations are information received from the environment, such as visual features or sounds. Perceptions are organized and interpreted sensations, such as recognizing a face or interpreting a sequence of sounds as familiar songs (p. 774). Podcasting relates to the four senses (hearing, smelling, tasting, and touching). Listening is the hearing of audio/video podcasts and controlling the speed of “listen and learn” is touching. Cognitive developmental theorist Jean Piaget (1896-1980) argued that “the knowledge and skill underlying intermodal perception can only be gained through experiences that involve looking at, hearing, smelling, tasting, and touching the external world” (p. 778).

Adapting to these theories, environmental and technological changes are commonly adopted by society. The Internet and innovations in Web technologies have breathed life into the knowledge society. Innovation in mobile learning and podcasting allows people to communicate and exchange information in today’s social structure. The question remains: “What is the best innovative way to learn?” One answer is to find a suitable way to adapt to today’s reality of students and educators. In this context, education through podcasting is a new teaching/learning methodology for the

realities of today's learning styles in academia. The new generation of college students already has extensive experience with the Internet, Web-enabled devices, and mobile technologies; and these technologies are fast becoming a significant part of their daily lives (Gallerani, 2008, p. 19). Due to social change, people are more readily accepting move toward the Internet, the Web, and mobile technology-based teaching/ learning. The Internet and Web technologies are indispensable elements in today's society. *San Jose Mercury News* columnist Mike Cassidy (2009) wrote a column called "Lessons Learned from a Tech-Free Day." He asked teenagers from the local high school to refrain from using their mobile devices for 12 hours: no iPods, no cell phones, no PDAs, no video games, no DVDs, no TVs, no PCs. Even though the "tech-free day" was merely a gesture, it showed that new generations of students are enthusiastic about their mobile devices.

Building on the successes of the Internet, Web-enabled and mobile technologies offer a wide range of applications across disciplines including business, education, and entertainment. This phenomenon opens up unlimited possibilities for how people live, learn, and work (Sorrentino, 2008b, pp. 215-216). It is a common behavior to learn and adapt to the information society, and the mobile society is a natural transition (Merriam & Cafarella, 1999, pp. 73-77). Consequently, higher education institutions and global industries need to form partnerships to enhance students' learning styles. Bill Gates (1999), founder of Microsoft, the world's most successful computer company, stated, "The Web lifestyle isn't about changing human nature or the fundamentals of how people live. Instead, the Web lifestyle gives people a chance to pursue their interests in a better way" (p. 131). Emerging mobile and podcasting technologies are just a natural evolution of "Web lifestyle" and beyond. The theory of knowing and the knowledge society aims to identify the most

appropriate ways to learn and succeed. This essentially means that, if the use of podcasting helps students in cultivating reason in the “real world,” then podcasting has achieved its purpose.

Mobile technologies have changed greatly since their inception. Today’s mobile computing can take the form of a personal computer, PDA, wireless network, cell phone, iPod, home network appliance, or any other digital electronic device. These digital devices are becoming an indispensable part of people’s lives, and they are responsible for connecting schools, businesses, and people from virtually anywhere in the world. Societal changes force people to apply learning in a more pragmatic way in education, and this is essentially what podcasting entails.

In terms of the value of mobile and podcasting technology to society, podcasting provides numerous ways for students to learn, and in effect, transforms classroom learning as we know it today. According to John Dewey (1859-1952), an American pragmatic philosopher and proponent of experimentalism in education, students learn by engaging in play, construction, nature study, and self-expression (Gutek, 1997, p. 82). Dewey’s school and classroom approaches were conceived as an embryonic community in which students work together to solve shared problems (p. 96).

Podcasting is also consistent with cooperative learning and Deutsch’s work in social psychology, which suggests that the social and motivational environment promotes positive, supportive peer interactions and a positive orientation toward achievement and learning (Salkind & Rasmussen, 2008, p. 188). If the effective use of podcasting improves students’ classroom experiences, then there is value added to the field of education, especially the teaching and learning environments. Based on what is known about podcasting, it can be an effective tool for facilitating collaboration and

information exchange. The podcasting learning concept is very similar to cooperative learning, which suggests that the social and motivational environment in the classroom promotes positive and supportive peer interactions and a positive orientation toward achievement and learning (Salkind & Rasmussen, pp. 187-188).

Podcasting, in learning theory and practice, is simply a concept of “listen and learn.” These educational practices provide opportunities for today’s lifestyles and learning styles of students. This pedagogical approach is consistent with the constructivist approach to self-directed learning that suggests that people construct new knowledge from their past experiences and that the curriculum and instruction should be tailored to facilitate development of individual perfections (Ferri, 2008, p. 82; Merriam & Caffarella, 1999, p. 69; Wong & Kerr, 2008, p. 42). Emerging podcasting methods for teaching and learning are based on the idea of anytime, anyplace learning to suit today’s lifestyles.

In terms of learning styles, learners learn best in their environment, namely learning styles and learning settings. According to Salkind and Rasmussen (2008), teachers should use learning style information to plan teaching strategies, learning activities, evaluation of learning, and development of programs and curricula (p. 598). According to Ned Herrmann (1922, as cited in Salkind & Rasmussen, pp. 600-601), there are four learning styles: (a) theorists (rational), (b) organizers (safe keeping), (c) innovators (experimental), and (d) humanitarians (feelings). Podcasting learning theory and practices are based on the idea of flexible learning styles suited to today’s students’ lifestyles.

Podcasting learning works like the traditional form of taking notes and reviewing classroom lecture notes. Traditionally, taking notes is a common activity in a classroom setting, and a large percentage of students are regular note takers

(Hussain, 2008, p. 69; Palmatier & Bennett, 1974). Several studies have investigated the art of effective note taking. According to Cohn, Cohn, and Bradley (1995), students must actively pay attention to the lecturer while listening and taking notes and then hold and manipulate a representation of that information in working memory while recording notes. As a consequence, students are less likely to pay attention to the lecture while taking notes. In another study, Kreiner (1997) found that interactive videotapes were more effective than taking notes while watching videotapes because they allowed students to focus on relevant concepts and encouraged classroom participation. Similarly, podcasting offers flexible, interactive learning outside the classroom and allows students to focus on the lecture.

Kiewra et al. (1991) outlined suggestions to facilitate selecting information to record during a lecture. Cueing strategies, such as recording information on the blackboard, repeating the lecture, and use of a tape or video recording, can be used repeatedly to enhance student learning. Classroom podcasts are essentially audio/video recordings of class notes, lecture notes, and cue cards that provide detailed summaries, outlines, or tutorials of the lessons. Fryer (2008) found that technology-assisted learning had a positive effect on lectures, particularly on notetaking and exam preparation. Smith and Woody (2000) found that technology-facilitated instruction was perceived more favorably by students than were traditional classroom instructions. Using podcasting technology to create highly participative, rich learning experiences, review classroom learning materials by reinforcing students' rapt attention, and respond more quickly to changes in environments enabled learning, fostering retention and more effective learning. Based on these findings, a technology-mediated approach significantly improved the student learning process and outcomes, as measured by student self-reports and objective outcome measures. A survey

conducted by the Berenson Strategy Group for Common Sense Media (2009) found that 41% of students (7th- to 12th-graders) reported storing class notes on their cell phones.

With educational podcasting, the Internet, and Web technologies, information-rich teaching and learning environments have been created that enable social interaction and the exchange of information between teachers and students. The Internet and Web technologies have enabled teachers and students to access a wealth of knowledge. Podcasting-supplemented learning tools can be used to replicate and expand the traditional classroom experience. Similarly, podcasting can be used to broaden access to, retrieve, and disseminate knowledge by providing anytime, anyplace learning (Aggarwal & Bento, 2000).

Today, technology is used in virtually all-American schools (K-12, colleges, and universities). Technology is integrated into the core classroom curriculum, making learning more accessible. By reaching out to the world through the Internet and now through mobile (m-learning), the experience can be extended beyond traditional forms of instruction (Lastrucci, Infante, & Pascale, 2008, p. 160). By weaving popular Internet communication services and students' technological culture, academic institutions may be able to engage and motivate students to learn.

The emerging mobile and portable technologies in cell phones, PDAs, and MP3 devices make podcasting possible. With podcasting, classroom instructional materials can be broadcast as digital audio/video learning modules that students can download to learn the material. Today's podcast has become ubiquitous for describing the packaging and distribution of audio/video content over the Internet. A definition of podcasting offered by Merriam-Webster is "a program (as of music or talk) made available in digital format for automatic download over the Internet" (Merriam-

Webster, 2001, p. 1). Innovations in podcasting have significantly changed teaching and learning in college education. With podcasting, the Internet, mobile and portable media devices, classroom instruction can be broadcast in RSS feed format and downloaded on demand. Like the Internet, mobile and portable media technologies provide greater flexibility in the learning process and are revitalizing today's college education (Reid-Martinez et al., 2008, p. 703).

Emerging podcasting technology enables community college students to access classroom learning materials on demand. This learning technique simply incorporates classroom experiences into students' own culture of technological identity to motivate them to learn the subject matter. Podcasting will not replace traditional classroom learning experiences but will instead reinforce students' appetite for learning. Several researchers have suggested that when students are motivated to learn in their technological identity environments, they are more eager to learn the subject matter, perform better on tests, and ultimately achieve better grades (Ranieri, 2008, p. 318). A study at the Catholic University Faculty of Economics showed that technology-mediated learning significantly improved students' achievement scores in mathematics tests (Torriero & Scancarello, 2008, p. 263). Dettori and Persico (2008) found that motivation was a key element in students' academic performance (p. 741). As O'Banion (1997) stated, "One can no longer afford to ignore technology and still maintain institutional health" (p. 64). Podcasting has become an indispensable part of the students' technological identity.

These technology trends will continue to provide new methods to help people interact, businesses extend commerce, and governments empower citizens and advance societal virtues. Based on technological trends, there is potential for convergence among business, education, and technology in knowledge sharing. The

vision of the future of technology will likely be based on centralized, knowledge-based systems that link knowledge databases. According to Motwani and Akbulut (2008), there will be more virtual classrooms in the future of educational training (p. 305). Given these results, growing numbers of teachers have already begun using technology-based media outlets and related technology-enabled instruction in their classroom learning environments.

Given the findings from these studies, educators will have a better understanding of students' use of technology relative to traditional classroom teaching methods. Educational podcasts are a relatively new technology that enable online audio/video communication between students and teachers, delivered via RSS and available for students to listen to on their PDAs or personal computers. In lieu of reviewing cryptic class notes, students can replay class lectures via the podcasting process. This also comes at a cost for both students and teachers, as they must have iPods, iPhones, or related MP3/MP4 players. In addition, teachers must regularly publish podcasts of the class materials.

In conclusion, there are many reasons for adapting to the real world of technological and societal changes. The general idea is to adapt, create, educate, empower, and innovate ways to enhance students' learning outcomes. By adapting to mainstream culture, technological changes, lifestyles, and learning styles of students, higher educational institutions can confront these changes by revitalizing "the old schools" thinking and create new ones that are in line with the real world of technological changes and societal changes (Goethals, Sorenson, & Burns, 2004, pp. 1134-1144).

In order to integrate podcasting effectively in the classroom teaching/learning environment, support effective training courses, and make the learning more effective,

future leaders of higher education must have a plan for societal changes and be ready to respond to technological challenges, especially with the younger generation of students who have grown up in the digital world. They must identify, recruit, and train staff members for leadership positions that help them to build more integrated ways to teaching and learning that will empower students to take control of their own learning and use the learning tools available to achieve desired learning outcomes. That includes training teachers to use technology to enhance their teaching methodology, support their personal development, and become more efficient and effective teachers. The technology trends discussed have shown that not only business enterprises, but also higher education institutions have been greatly affected by the advent of the Internet and, more recently, by mobile technologies such as cell phones, PDAs, and related media devices. These innovations have virtually transformed every aspect of society, including higher education, business, government, and global industries. Many of these institutions have already undergone paradigm shifts and major repercussions in restructuring to meet the needs of all their stakeholders.

For these reasons, future leaders in higher education must find new ways to overcome the digital divide between teachers and students, particularly those who are academically underprepared and economically disadvantaged. In addition, they must put their stakeholders' needs at the forefront and adapt to societal and technological changes to reach future students. After all, the real purpose of education is to educate students. The challenge is not so much about podcasting and its ability to package/distribute course content as digital content; rather, it is to ensure that quality teaching meets students' needs and satisfies learning requirements by embracing societal and technological changes without derailing the institution's purpose.

Podcasting is a new way of teaching and interaction between learners and knowledge. As Bill Gates (1999) pointed out, “We always overestimate the change that will occur in two years and underestimate the change that will occur in the next ten” (p. 69). For these reasons, understanding the role of students' technological culture and the technological society is a must for future leaders in higher education.

WSIS acknowledged that it is a global responsibility, including higher education, to “create access, utilize and share information and knowledge, enabling individuals, communities and peoples to achieve their full potential” (WSIS, 2003, para. 1). For this reason, higher institutions of learning must reconcile the importance of technology-based teaching/learning to the overall success of the institutions' practices.